

Cisco N9300-FX2 Series Switches



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Product overview

The Cisco N9300-FX2 Series switches belongs to the fixed Cisco N9000 platform based on [Cisco Cloud Scale technology](#). The platform support cost-effective cloud-scale deployments, an increased number of endpoints, and cloud services with wire-rate security and telemetry. The platform is built on modern system architecture designed to provide high performance and meet the evolving needs of highly scalable data centers and growing enterprises.

Cisco N9300-FX2 series is an extension of N9300-FX series switches with higher bandwidth capacity. The switches offer a variety of interface options to transparently migrate existing data centers from 1-Gbps, and 10-Gbps speeds to 25- Gbps at the server, and from 10- and 40-Gbps speeds to 50- and 100- Gbps at the aggregation layer. The platforms provide investment protection for customers, delivering large buffers, highly flexible Layer 2 and Layer 3 scalability, and performance to meet the changing needs of virtualized data centers and automated cloud environments.

Cisco provides two modes of operation for Cisco N9000 Series Switches. Organizations can use [Cisco NX-OS Software](#) to deploy the switches in standard Cisco Nexus switch environments (NX-OS mode). Organizations can also deploy the infrastructure that is ready to support the [Cisco Application Centric Infrastructure](#) (Cisco ACI™) platform to take full advantage of an automated, policy-based, systems-management approach (ACI mode).

Switch models

Table 1. Cisco N9300-FX2 Series Switches

Model	Description
Cisco N9336C-FX2	36 x 40/100-Gbps QSFP28 ports
Cisco N9336C-FX2-E	36 x 40/100-Gbps QSFP28 ports
Cisco N93240YC-FX2	48 x 1/10/25-Gbps fiber ports and 12 x 40/100-Gbps QSFP28 ports
Cisco N93360YC-FX2	96 x 1/10/25-Gbps fiber ports and 12 x 40/100-Gbps QSFP28 ports
Cisco N93216TC-FX2	96 x 100M/1/10GBASE-T ports and 12 x 40/100-Gigabit QSFP28 ports

The Cisco N9336C-FX2 Switch (Figure 1) is a 1RU switch that supports 7.2 Tbps of bandwidth and over 2.4 bpps. The switch can be configured to work as 1/10/25/40/50/100-Gbps offering flexible options in a compact form factor. Breakout is supported on all ports. Please see feature table below for more information.



Figure 1.
Cisco N9336C-FX2 Switch

The Cisco N9336C-FX2-E Switch (Figure 2) is a 1RU switch that supports 7.2 Tbps of bandwidth and over 2.4 bpps. The switch can be configured to work as 1/10/25/40/50/100-Gbps or as 16-, 32-Gbps Fibre Channel ports^[2] offering flexible options in a compact form factor. Breakout is supported on all ports. Please see feature table below for more information.



Figure 2.
Cisco N9336C-FX2-E Switch

The Cisco N93240YC-FX2 Switch (Figure 3) is a 1.2RU switch that supports 4.8 Tbps of bandwidth and over 2.5bps. The 48 ports of downlinks support 1/10/25-Gbps. The 12 uplinks ports can be configured as 40- and 100-Gbps ports, offering flexible migration options. The switch is ideal for a non-oversubscribed solution in a compact form factor. The switch has FC-FEC and RS-FEC enabled for 25Gbps support over longer distances.



Figure 3.
Cisco N93240YC-FX2 Switch

The Cisco N93360YC-FX2 Leaf Switch is an 2-Rack-Unit (2RU) Leaf switch that supports 7.2 Tbps of bandwidth and 2.4 bpps across 96 fixed 10/25G SFP+ ports and 12 fixed 40/100G QSFP28 ports (Figure 3). The 96 ports of downlinks support 1/10/25-Gbps. The 12 uplinks ports can be configured as 40- and 100-Gbps ports, offering flexible migration options. The switch has FC-FEC and RS-FEC enabled for 25Gbps support over longer distances. Please see feature table below for more information.



Figure 4.
Cisco N93360YC-FX2 Switch

The Cisco N93216TC-FX2 Switch (Figure 5) is a 2RU switch that supports 4.32 Tbps of bandwidth and over 2.5bps. The 96 10GBASE-T downlink ports on the 93216TC-FX2 can be configured to work as 100-Mbps, 1-Gbps, or 10-Gbps ports. The 12 uplinks ports can be configured as 40- and 100-Gbps ports, offering flexible migration options.



Figure 5.
Cisco N93216TC-FX2 Switch

Features and benefits

The Cisco N9300-FX2 series provide the following features and benefits:

- **Architectural Flexibility**
 - Industry leading Software Defined Networking Solution Cisco ACI™ support.
 - Support for standards based VXLAN EVPN fabrics, inclusive of hierarchical multi-site support (refer to [VXLAN Network with MP-BGP EVPN Control Plane](#) for more information).
 - Three-tier BGP architectures, enabling horizontal, non-blocking IPv6 network fabrics at web-scale.
 - Segment routing allows the network to forward Multiprotocol Label Switching (MPLS) packets and engineer traffic without Resource Reservation Protocol (RSVP) Traffic Engineering (TE). It provides a control-plane alternative for increased network scalability and virtualization.
 - Comprehensive protocol support for Layer 3 (v4/v6) unicast and multicast routing protocol suites, including BGP, Open Shortest Path First (OSPF), Interior System to Interior System (IS-IS), Enhanced Interior Gateway Routing Protocol (EIGRP), Routing Information Protocol Version 2 (RIPv2), Protocol Independent Multicast Sparse Mode (PIM-SM), Source-Specific Multicast (SSM), and Multicast Source Discovery Protocol (MSDP).
- **Extensive Programmability**
 - Day zero automation through Power On Auto Provisioning, drastically reducing provisioning time.
 - Industry leading integrations for leading develops configuration management applications – Ansible, Chef, Puppet, SALT. Extensive Native YANG and industry standard OpenConfig model support through RESTCONF/NETCONF.
 - Pervasive API's for all switch CLI functions (JSON based RPC over HTTP/HTTPS).

- **High Scalability, flexibility, and security**

- Flexible forwarding tables support up to 1 million shared entries on FX2 models. Flexible use of TCAM space allows for custom definition of Access Control List (ACL) templates.
- IEEE 802.1ae MAC Security (MACsec¹) support on all ports of 9300-FX2 models with speed greater than or equal to 1-Gbps, allows traffic encryption at the physical layer and provides secure server, border leaf, and leaf-to-spine connectivity.

- **Intelligent Buffer Management**

- The platform offers Cisco's innovative [intelligent buffer management](#), that offers capability to distinguish mice and elephant flows and apply different queue management schemes to them based on their network forwarding requirements in the event of link congestion.
- Intelligent buffer management functions are:
 - Approximate Fair Dropping (AFD) with Elephant Trap (ETRAP). AFD distinguishes long-lived elephant flows from short-lived mice flows, by using ETRAP. AFD exempts mice flows from the dropping algorithm so that mice flows will get their fair share of bandwidth without being starved by bandwidth-hungry elephant flows. Also, AFD tracks elephant flows and subjects them to the AFD algorithm in the egress queue to grant them their fair share of bandwidth.
 - ETRAP measures the byte counts of incoming flows and compares this against the user defined ETRAP threshold. After a flow crosses the threshold, it becomes an elephant flow.
 - Dynamic Packet Prioritization (DPP) provides the capability of separating mice flows and elephant flows into two different queues so that buffer space can be allocated to them independently. Mice flows, sensitive to congestion and latency can take priority queue and avoid re-ordering that allows to elephant flows to take full link bandwidth.

- **RDMA over Converged Ethernet – RoCE support**

- Platform offers lossless transport for RDMA over Converged Ethernet with support of DCB protocols:
 - Priority-based Flow Control – (PFC) to prevent drops in the network and pause frame propagation per priority class.
 - Enhanced Transmission Selection – (ETS) to reserve bandwidth per priority class in network contention situation.
 - Data Center Bridging Exchange Protocol – (DCBX) to discover and exchange priority and bandwidth information with end points.
- Platform also supports Explicit Congestion Notification – (ECN) that provides end-to-end notification per IP flow by marking packets that experienced congestion, without dropping traffic. The platform is capable to track ECN statistics of number of marked packet that have experienced congestion.

¹ Please check software release notes to get the latest support for each product to enable MACsec.

- **LAN and SAN Convergence**

- Fibre Channel and Fibre Channel over Ethernet (FCoE) N-Port Virtualization (NPV) support enables the network administrator to control domain IDs and points of management on a Fibre Channel network as it scales. This feature enables LAN and SAN converged networks on a lossless, reliable Ethernet network.

- **Hardware and software high availability**

- Virtual Port-Channel (vPC) technology provides Layer 2 multipathing through the elimination of Spanning Tree Protocol. It also enables fully utilized bisectional bandwidth and simplified Layer 2 logical topologies without the need to change the existing management and deployment models.
- The 64-way Equal-Cost MultiPath (ECMP) routing enables the use of Layer 3 fat-tree designs. This feature helps organizations prevent network bottlenecks, increase resiliency, and add capacity with little network disruption.
- Advanced reboot capabilities include hot and cold patching.
- The switches use hot-swappable Power-Supply Units (PSUs) and fans with N+1 redundancy.

- **Purpose-built Cisco NX-OS Software operating system with comprehensive, proven innovations**

- Single binary image that supports every switch in the Cisco N9000 series, simplifying image management. The operating system is modular, with a dedicated process for each routing protocol: a design that isolates faults while increasing availability. In the event of a process failure, the process can be restarted without loss of state. The operating system supports hot and cold patching and online diagnostics.
- Data Center Network Manager (DCNM) is the network management platform for all NX-OS-enabled deployments, spanning new fabric architectures, IP Fabric for Media, and storage networking deployments for the Cisco Nexus®-powered data center. Accelerate provisioning from days to minutes, and simplify deployments from day zero through day N. Reduce troubleshooting cycles with graphical operational visibility for topology, network fabric, and infrastructure. Eliminate configuration errors and automate ongoing change in a closed loop, with templated deployment models and configuration compliance alerting with automatic remediation. Real-time health summary for fabric, devices, and topology. Correlated visibility for fabric (underlay, overlay, virtual and physical endpoints), including compute visualization with VMware.
- Network traffic monitoring with Cisco Nexus Data Broker builds simple, scalable, and cost-effective network Test Access Points (TAPs) and Cisco Switched Port Analyzer (SPAN) aggregation for network traffic monitoring and analysis.

- **Cisco Tetration Analytics platform support**

- The telemetry information from the N9300 Series switches is exported every 100 milliseconds by default directly from the switch's Application-Specific Integrated Circuit (ASIC). This information consists of three types of data: (a) Flow information, this information contains information about endpoints, protocols, ports, when the flow started, how long the flow was active, etc. (b) Inter-packet variation, this information captures any inter-packet variations within the flow. Examples include variation in Time To Live (TTL), IP and TCP flags, payload length, etc. (c) Context details, context information is derived outside the packet header, including variation in buffer utilization, packet drops within a flow, association with tunnel endpoints, etc.
- The Cisco Tetration Analytics platform consumes this telemetry data, and by using unsupervised machine learning and behavior analysis it can provide outstanding pervasive visibility across everything in your data center in real time. By using algorithmic approaches, the Cisco Tetration Analytics platform provides a deep application insights and interactions, enabling dramatically simplified operations, a zero-trust model, and migration of applications to any programmable infrastructure. To learn more, go to <https://www.cisco.com/go/tetration>.

- **Cisco Network Assurance Engine (NAE)**

- Cisco NAE continuously verifies if the network infrastructure is operating as per policy intent and it leverages the power of mathematical models to reason on behalf of the operator in policy, configuration and dynamic state level. NAE can precisely indicate problems in the network, identify which application or part of network is impacted, root-cause the problem and suggest how to fix it. Its continuous verification approach transforms Day 2 Operations from reactive to proactive mode and it does so without using any packet data. NAE helps avoid outages by predicting the impact of changes, reducing network related IT incidents and shrinking the mean time to repair by up to 66%. NAE also helps assure network security and segmentation compliance. To learn more about NAE, visit <https://www.cisco.com/c/en/us/products/data-center-analytics/network-assurance-engine/index.html>.

Product specifications

The Cisco N9300-FX2 series offer industry-leading density and performance with flexible port configurations that can support existing copper and fiber cabling (Table 2).

Table 2. Cisco N9300-FX2 Series Switches specifications

Feature	Cisco N9336C-FX2	Cisco N9336C-FX2-E	Cisco N93240YC-FX2	Cisco N93360YC-FX2	Cisco N93216TC-FX2
Ports	36 x 40/100-Gbps QSFP28 ports	36 x 40/100-Gbps QSFP28 ports	48 x 1/10/25-Gbps and 12 x 40/100-Gbps QSFP28 ports	96 x 1/10/25-Gbps and 12 x 40/100-Gbps QSFP28 ports	96 x 100M/1/10GBASE-T ports and 12 x 40/100-Gbps QSFP28 ports
Supported speeds	1/10/25/40/100-Gbps Ethernet Breakout supported on all ports, 1-36: 100G, 2x50G NRZ, 40G native, 4x10/25G (10G w/QSA) 1G w/QSA except ports 1-6 and 33-36	1/10/25/40/100-Gbps Ethernet Breakout supported on all ports, 1-36: 100G, 2x50G NRZ, 40G native, 4x10/25G (10G w/QSA) 1G w/QSA except ports 1-6 and 33-36	1/10/25-Gbps on downlinks 40/100-Gbps on uplinks	1/10/25-Gbps on downlinks 40/100-Gbps on uplinks Breakout supported ports, 97-108: 4x10/25G	100M/1/10 Gbps RJ45 downlinks 40/100-Gbps on uplinks Breakout supported ports, 97-108: 4x10/25G
FC/FCoE	N/A	16/32-Gbps Fiber Channel	N/A	16/32-Gbps Fiber Channel	N/A
CPU	4 cores	4 cores	4 cores	4 cores	4 cores
System memory	24 GB	24 GB	Upto 24 GB	Upto 24 GB	Upto 24 GB
SSD drive	128 GB	128 GB	128 GB	128 GB	128 GB
System buffer	40 MB	40 MB	40 MB	40 MB	40 MB
Management ports	2 ports: 1 RJ-45 and 1 SFP+	2 ports: 1 RJ-45 and 1 SFP+	2 ports: 1 RJ-45 and 1 SFP+	2 ports: 1 RJ-45 and 1 SFP+	2 ports: 1 RJ-45 and 1 SFP+
USB ports	1	1	1	1	1
RS-232 serial ports	1	1	1	1	1

Feature	Cisco N9336C-FX2	Cisco N9336C-FX2-E	Cisco N93240YC-FX2	Cisco N93360YC-FX2	Cisco N93216TC-FX2
Power supplies (up to 2)	750W AC ² , 1100W AC, 1100W DC, 1100W HVAC/HVDC	1100W AC, 1100W DC, 1100W HVAC/HVDC	750W AC ² , 1100W AC, 1100W DC, 1100W HVAC/HVDC	1200W AC, 930W DC, 1200W HVAC/HVDC	1200W AC, 930W DC, 1200W HVAC/HVDC
Typical power (AC)	337W	337W	298W	404W	580W
Maximum power (AC)	719W	705W	708W	900W	965W
Input voltage (AC)	100 to 240V	100 to 240V	100 to 240V	100 to 240V	100 to 240V
Input voltage (High-Voltage AC [HVAC])	100 to 277V	100 to 277V	100 to 277V	100 to 277V	100 to 277V
Input voltage (DC)	-40 to -72V	-40 to -72V	-40 to -72V	-40V to -72V	-40V to -72V
Input voltage (High-Voltage DC [HVDC])	-240 to -380V	-240 to -380V	-240 to -380V	-240V to -380V	-240V to -380V
Frequency (AC)	50 to 60 Hz	50 to 60 Hz	50 to 60 Hz	50 to 60 Hz	50 to 60 Hz
Fans	3 dual fan trays	6	5	3 fan trays	3 fan trays
Airflow	Port-side intake and exhaust	Port-side intake and exhaust	Port-side intake and exhaust	Port-side intake and exhaust	Port-side intake and exhaust
Physical dimensions (H x W x D)	1.72 x 17.3 x 24.5 in. (4.4 x 43.9 x 62.3 cm)	1.72 x 17.3 x 24.7 in. (4.4 x 43.9 x 62.7 cm)	2.1 x 17.3 x 23.3 in. (5.3 x 43.9 x 59.1 cm)	3.38 x 17.41 x 24.14 in. (8.59 x 44.23 x 61.31 cm)	3.38 x 17.41 x 23.6 in (8.59 x 44.2 x 59.9 cm)
Acoustics	76.2 dBA at 50% fan speed, 85.3 dBA at 70% fan speed, and 92.3 dBA at 100% fan speed	74.7 dBA at 50% fan speed, 80.4 dBA at 70% fan speed, and 88.1 dBA at 100% fan speed	76.4 dBA at 50% fan speed, 83.3 dBA at 70% fan speed, and 92.1 dBA at 100% fan speed	76.7 dBA at 40% fan speed, 88.7 dBA at 70% Fan speed and 97.4 dBA at 100% Fan speed	76.7 dBA at 40% fan speed, 88.7 dBA at 70% Fan speed and 97.4 dBA at 100% Fan speed
RoHS compliance	Yes	Yes	Yes	Yes	Yes
MTBF	335,900 hours	330,810 hours	345,950 hours	318,030 hours	267,560 hours
Minimum ACI image	ACI-N9KDK9-13.1.2		ACI-N9KDK9-14.0	ACI-N9KDK9-14.1.2	ACI-N9KDK9-14.1.2
Minimum NX-OS image	NXOS-703I7.3	NXOS-10.1(1)	NXOS-703I7.3	NXOS-9.3(1)	NXOS-9.3(1)

² Typical and maximum power values are based on input drawn from the power circuit. The power supply value (for example, 750W AC power supply: NXA-PAC-750W-PI) is based on the output rating to the inside of the switch.

Table 3 lists the performance and scalability specifications for the Cisco N9300-FX2/FX2-E series switches. (Check the software release notes for feature support information.)

Table 3. Hardware performance and scalability specifications³

Item	Cisco N9300-FX2 Series Switches
Maximum number of IPv4 Longest Prefix Match (LPM) routes ⁴	896,000
Maximum number of IPv4 host entries ⁴	896,000
Maximum number of IPv6 Longest Prefix Match (LPM) routes ⁴	498,000
Maximum number of IPv6 host entries ⁴	896,000
Maximum number of MAC address entries ⁴	256,000
Maximum number of multicast routes	128,000
Number of Internet Group Management Protocol (IGMP) snooping groups	Shipping: 8,000 Maximum: 32,000
Maximum number of Cisco N2000 Series Fabric Extenders per switch	16
Maximum number of Access Control List (ACL) entries	Per slice of the forwarding engine: 5000 ingress 2000 egress
Maximum number of VLANs	4096 ⁵
Number of Virtual Routing and Forwarding (VRF) instances	Shipping: 1,000 Maximum: 16,000
Maximum number of ECMP paths	64
Maximum number of port channels	512
Maximum number of links in a port channel	32
Number of active SPAN sessions	4
Maximum number of VLAN's in Rapid per-VLAN Spanning Tree (RPVST) instances	3,967
Maximum number of Hot-Standby Router Protocol (HSRP) groups	490
Number of Network Address Translation (NAT) entries	1,023
Maximum number of Multiple Spanning Tree (MST) instances	64

Item	Cisco N9300-FX2 Series Switches
Flow-table size used for Cisco Tetration Analytics platform	64,000
Number of Queues	8

³ More templates and greater scalability are on the roadmap. Refer to the [Cisco N9000 Series Verified Scalability Guide](#) documentation for the latest exact scalability values validated for specific software.

⁴ Raw capacity of tables.

⁵ 127 VLANs out of 4096 are reserved.

Table 4 lists the environmental properties, and Table 5 lists the weight for the Cisco N9300-FX2 series switches.

Table 4. Environmental properties

Property	Description
Operating temperature	32 to 104° F (0 to 40° C)
Nonoperating (storage) temperature	–40 to 158° F (–40 to 70° C)
Humidity	5 to 95% (noncondensing)
Altitude	0 to 13,123 ft (0 to 4000m)

Table 5. Weight

Component	Weight
Cisco N9336C-FX2 without power supplies or fans	18.8 lb (8.5 kg)
Cisco N9336C-FX2-E Chassis weight per unit	18.8 lb (8.5 kg)
Cisco N93240YC-FX2 without power supplies or fans	2.2 lb (10 kg)
750W* AC power supply	2.42 lb (1.1 kg)
1100W AC power supply	2.42 lb (1.1 kg)
1100W DC power supply	2.45 lb (1.11 kg)
1100W HVAC/HVDC power supply	2.46 lb (1.12 kg)
Cisco N93360YC-FX2 without power supplies or fans	27.4 lb (12.4kg)
Cisco N93216TC-FX2 without power supplies or fans	27.4 lb (12.4 kg)
930W DC Power Supply	2.42 lb (1.1 kg)
1200W AC power supply	2.64 lb (1.2 kg)
1200W HVAC/HVDC power supply	2.52 lb (1.14kg)

Component	Weight
Fan tray: NXA-FAN-65CFM-F or NXA-FAN-65CFM-B	0.6 lb (0.3 kg)
Fan tray: NXA-FAN-35CFM-PE or NXA-FAN-35CFM-PI	0.26 lb (0.1 kg)
NXA-FAN-160CFM-PI or NXA-FAN-160CFM-PE	1.3 lb (0.59 kg)

Table 6 summarizes regulatory standards compliance for the Cisco N9300-FX2 series switches.

Table 6. Regulatory standards compliance: Safety and EMC

Specification	Description
Regulatory compliance	Products should comply with CE Markings according to directives 2004/108/EC and 2006/95/EC
Safety	NEBS • UL 60950-1 Second Edition • CAN/CSA-C22.2 No. 60950-1 Second Edition • EN 60950-1 Second Edition • IEC 60950-1 Second Edition • AS/NZS 60950-1 • GB4943
EMC: Emissions	• 47CFR Part 15 (CFR 47) Class A • AS/NZS CISPR22 Class A • CISPR22 Class A • EN55022 Class A • ICES003 Class A • VCCI Class A • EN61000-3-2 • EN61000-3-3 • KN22 Class A • CNS13438 Class A
EMC: Immunity	• EN55024 • CISPR24 • EN300386 • KN 61000-4 series
RoHS	The product is RoHS-6 compliant with exceptions for leaded-Ball Grid-Array (BGA) balls and lead press-fit connectors.

Software licensing and optics supported

The software packaging for the Cisco N9000 Series offers flexibility and a comprehensive feature set. The default system software has a comprehensive Layer 2 security and management feature set. To enable additional functions, including Layer 3 IP unicast and IP multicast routing and Cisco Nexus Data Broker, you must install additional licenses. To meet customer requirements, licensing is available as both subscription and perpetual. The [licensing guide](#) illustrates the software packaging and licensing available to enable advanced features. For the latest software release information and recommendations, refer to the product bulletin at <https://www.cisco.com/go/nexus9000>.

For details about the optics modules available and the minimum software release required for each supported module, visit https://www.cisco.com/en/US/products/hw/modules/ps5455/products_device_support_tables_list.html.

Ordering information

Table 7 presents ordering information for the Cisco N9300-FX series switches.

Table 7. Ordering information

Part Number	Product Description
Base Part Numbers	
N9K-C9336C-FX2	Nexus 9K Fixed with 36p 40G/100G QSFP28
N9K-C9336C-FX2-E	Nexus 9K Fixed with 36p 40G/100G QSFP28
N9K-C93240YC-FX2	Nexus 9K Fixed with 48p 1/10G/25G SFP and 12p 40G/100G QSFP28
N9K-C93360YC-FX2	Nexus 9K Fixed with 96p 1/10G/25G SFP and 12p 40G/100G QSFP28
N9K-C93216TC-FX2	96p 100M/1/10GBASE-T and 12p 40G/100G QSFP28
Power Supplies on N9300-FX2 series	
NXA-PAC-750W-PI*	N9000 750W AC PS, Port-side Intake
NXA-PAC-750W-PE*	N9000 750W AC PS, Port-side Exhaust
NXA-PAC-1100W-PI2	N9000 1100W AC PS, Port-side Intake
NXA-PAC-1100W-PE2	N9000 1100W AC PS, Port-side Exhaust
NXA-PDC-1100W-PI	N9000 1100W DC PS, Port-side Intake
NXA-PDC-1100W-PE	N9000 1100W DC PS, Port-side Exhaust
NXA-PHV-1100W-PI	N1100W Platinum HV-AC-DC PS, Port-side Intake
NXA-PHV-1100W-PE	N1100W Platinum HV-AC-DC PS, Port-side Exhaust
NXA-PAC-1200W-PE	Cisco N1200W AC PS, Port-side Exhaust
NXA-PAC-1200W-PI	Cisco N1200W AC PS, Port-side Intake

Part Number	Product Description
N9K-PUV-1200W	Cisco N1200W, 200-277AC,240-380DC, dual airflow PSU
NXA-PDC-930W-PI	Cisco N930W DC PS, Port-side Intake
NXA-PDC-930W-PE	Cisco N930W DC PS, Port-side Exhaust
Fans on N9300-FX2 Series	
NXA-FAN-35CFM-PE	Nexus Single Fan, 35CFM, port side exhaust airflow; supported on N93240YC-FX2
NXA-FAN-35CFM-PI	Nexus Single Fan, 35CFM, port side intake airflow; supported on N93240YC-FX2
NXA-FAN-65CFM-PE	Nexus Dual Fan, 65CFM, port side exhaust airflow; supported on N9336C-FX2
NXA-FAN-65CFM-PI	Nexus Dual Fan, 65CFM, port side intake airflow; supported on N9336C-FX2
NXA-FAN-160CFM-PI	Cisco Nexus Fan, 160CFM, port-side intake airflow
NXA-FAN-160CFM-PE	Cisco Nexus Fan, 160CFM, port-side exhaust airflow
Licenses on N9300-FX2 Series	
C1E1TN9300XF-3Y	ACI & NX-OS Subscription Essential package for 10/25/40G+ Nexus 9K Leaf, 3 Year Term
C1E1TN9300XF-5Y	ACI & NX-OS Subscription Essential package for 10/25/40G+ Nexus 9K Leaf, 5 Year Term
C1A1TN9300XF-3Y	ACI & NX-OS Subscription Advantage package for 10/25/40G+ Nexus 9K Leaf, 3 Year Term
C1A1TN9300XF-5Y	ACI & NX-OS Subscription Advantage package for 10/25/40G+ Nexus 9K Leaf, 5 Year Term
ACI-ES-XF	ACI Essential SW license for a 10/25/40G+ Nexus 9K Leaf
ACI-AD-XF	ACI Advantage SW license for a 10/25/40G+ Nexus 9K Leaf
NX-OS-ES-XF	NX-OS Essential SW license for a 10/25/40G+ Nexus 9K Leaf
NX-OS-AD-XF	NX-OS Advantage SW license for a 10/25/40G+ Nexus 9K Leaf
Power Cords	
CAB-250V-10A-AR	AC Power Cord - 250V, 10A - Argentina (2.5 meter)
CAB-250V-10A-BR	AC Power Cord - 250V, 10A - Brazil (2.1 meter)
CAB-250V-10A-CN	AC Power Cord - 250V, 10A - PRC (2.5 meter)
CAB-250V-10A-ID	AC Power Cord - 250V, 10A, South Africa (2.5 meter)
CAB-250V-10A-IS	AC Power Cord - 250V, 10A - Israel (2.5 meter)
CAB-9K10A-AU	Power Cord, 250VAC 10A 3112 Plug, Australia (2.5 meter)
CAB-9K10A-EU	Power Cord, 250VAC 10A CEE 7/7 Plug, EU (2.5 meter)

Part Number	Product Description
CAB-9K10A-IT	Power Cord, 250VAC 10A CEI 23-16/VII Plug, Italy (2.5 meter)
CAB-9K10A-SW	Power Cord, 250VAC 10A MP232 Plug, SWITZ (2.5 meter)
CAB-9K10A-UK	Power Cord, 250VAC 10A BS1363 Plug (13 A fuse), UK (2.5 meter)
CAB-9K12A-NA	Power Cord, 125VAC 13A NEMA 5-15 Plug, North America (2.5 meter)
CAB-AC-L620-C13	North America, NEMA L6-20-C13 (2.0 meter)
CAB-C13-C14-2M	Power Cord Jumper, C13-C14 Connectors, 2 Meter Length (2 meter)
CAB-C13-CBN	Cabinet Jumper Power Cord, 250 VAC 10A, C14-C13 Connectors (0.7 meter)
CAB-IND-10A	10A Power cable for India (2.5 meter)
CAB-N5K6A-NA	Power Cord, 200/240V 6A North America (2.5 meter)
CAB-HVAC-SD-0.6M	HVAC Power cable for Anderson-LS-25
CAB-HVAC-RT-0.6M	HVAC Power cable with right angle connector for RF-LS-25
Accessories on N9300-FX2 Series	
NXK-ACC-KIT-1RU	Nexus Fixed Accessory Kit with 4-post rack mount kit

Warranty, service and support

The Cisco N9300-FX2 series has a 1-year limited hardware warranty. The warranty includes hardware replacement with a 10-day turnaround from receipt of a Return Materials Authorization (RMA).

Cisco offers a range of professional, solution, and product support services for each stage of your Cisco N9300-FX2 series deployment:

- Cisco Data Center Quick Start Service for Cisco N9000 Series Switches: This offering provides consulting services that include technical advice and assistance to help deploy Cisco N9000 Series Switches.
- Cisco Data Center Accelerated Deployment Service for Cisco N9000 Series Switches: This service delivers planning, design, and implementation expertise to bring your project into production. The service also provides recommended next steps, an architectural high-level design, and operation-readiness guidelines to scale the implementation to your environment.
- Cisco Migration Service for Cisco N9000 Series Switches: This service helps you migrate from Cisco Catalyst® 6000 Series Switches to Cisco N9000 Series Switches.

- Cisco Product Support: Support service is available globally 24 hours a day, 7 days a week, for Cisco software and hardware products and technologies associated with Cisco N9000 Series Switches. Enhanced support options delivered by Cisco also include solution support for Cisco ACI, Cisco SMARTnet™ Service, and Cisco Smart Net Total Care™ service.

For more information, visit <https://www.cisco.com/go/services>.

* For Cisco products only.

Cisco environmental sustainability

Information about Cisco’s environmental sustainability policies and initiatives for our products, solutions, operations, and extended operations or supply chain is provided in the “Environment Sustainability” section of Cisco’s 2018 [Corporate Social Responsibility](#) (CSR) Report.

Reference links to **information about key environmental sustainability topics** (mentioned in the “Environment Sustainability” section of the CSR Report) are provided in the following table:

Sustainability Topic	Reference
Information on product-material-content laws and regulations	Materials
Information on electronic waste laws and regulations, including products, batteries and packaging	WEEE Compliance

Reference links to **product-specific environmental sustainability information** that is mentioned in relevant sections of this data sheet are provided in the following table:

Sustainability Topic	Reference
General	
Product Compliance	Table 6. Safety and Compliance Information
Power	
Power Supply	Table 2. Product specifications: Power Supplies. Typical and Max power specification
Material	
Unit Weight	Table 5. Weight
Dimensions and Mean Time between Failures Metrics	Table 2. Product specifications

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For more information

For more information about the Cisco N9000 Series and latest software release information and recommendations, visit <https://www.cisco.com/go/nexus9000>.

Document history

New or revised topic	Described In	Date
Updated security information	Features and benefits	July 19, 2023
Product name change	Throughout the data sheet	August 2026

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